Audit, step 3, area B: layout and decoupling optimiser

Auditor: an independent agent (model: Fable), read-only, with scratch test scripts. HEAD 2734546. **No critical or high-severity bug found.** Fix status is in `DECISIONS.md` ("Audit actions").

ID	Severity	Status	Finding	Evidence
B1	Medium	CONFIRMED	In ring mode, `tableZero` is null, so every pair gets `ring1: false` and `kind: 'next'`. "Beyond nearest ring" then equals the adjacent set, `decoupledRing1` is always false, and the pair table labels adjacent pairs "next"	N=16 ring: 33 adjacent pairs labelled 'next'; next.max = adjacent.max
B2	Medium	CONFIRMED	Ring-mode `element` panel, `coreRadius` and `h` describe ring 0 only; `elements[]` has no `h`	N=24 ring: panel Rf 33.5 vs ring Rfs 33.5 / 37.0 / 38.5
B3	Low	CONFIRMED	Comment says a top ring of 1–2 becomes a cap; the code only treats n = 1 that way	N=6 → [2,4] with no cap
B4	Low	CONFIRMED	User `ringCounts` whose sum ≠ N silently replaces N	N=16 with [2,6,10] → N=18
B5	Low	LIKELY	`liftFeasible` set false permanently by any rejected trial; never re-evaluated for the final rings	Code reading
B6	Low	LIKELY	Free mode: shared Rf unbounded; lift solved once at Rf_o; clearance not re-checked at the final Rf	Benign in tests (lift saturates at c/2)
B7	Low	POSSIBLE	Ring-mode cost over a Voronoi set recomputed per trial, so the objective is discontinuous	No escape observed; adjacency stable at 0.5–2× tolerance
B8	Low	CONFIRMED (docs)	Design §5.1 "not yet implemented" is stale; §5.2 says "spherical spiral" (code: farthest-point sampling + Lloyd); duplicated zCut rows	grep
B9	Low	POSSIBLE	C2: free arm is best of 4 starts on the primary metric, ring arm a single start, so selection favours free (pre-registered that way)	Protocol
B10	Low	POSSIBLE	Two different pair-null estimators (free: tableZero; ring: last pairScan zero)	They agree to 3 decimals for 4 knots

Verified correct

- **Lookup angle convention:** worst error 1.0 nH on |M| up to 123 nH; the alternative convention would be off by up to 19 nH.
- **Index handling:** lookup wrapping and indices; ring twin indices and mirror flags for N = 2 to 64; count allocation always sums to N.
- **Caches:** keys are sufficient (no stale reuse).

- **Sample sets:** adjacency and coverage use consistent samples; adjacency is robust to tolerance ×0.5 and ×2.
- **Search moves:** zone-first acceptance, rescue jumps, size steps and locked-rotation steps are sound.
- **Multi-start:** reproducible; the chosen start re-runs exactly.
- **Verdict logic:** “decoupled” ⇔ all $|k| < 0.01$; verdicts are false, not vacuously true, when a category is empty.